Target-Specific ALIGNN Improvements for Band Gap and Energy Above Hull

ALIGNN Reimplementation Study

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1 Scope

This report documents only the two new JARVIS targets improved after the formation-energy study:

- `optb88vdw_bandgap`
- `ehull`

All comparisons used the same target, same dataset, same exact imported train/validation/test JIDs, and the same 1000 test structures as the original repository runs. The original comparison artifacts came from `$HOME/projects/2d-alignn`, while the improved reimplementation runs were executed under `$HOME/projects/alignn` on the remote GPU instance.

2 What Was Different From the Original Model

The original model used a generic graph-level regression setup across targets. Our improved runs kept the same ALIGNN-style architecture scale used in these comparisons, but changed the training objective to match each target's observed failure mode.

2.1 Shared Implementation Differences

The reimplementation included several fixes and training controls that were not part of the original comparison baseline:

- Original-style normalized edge-gated graph convolution with residual updates, batch normalization, and SiLU activations.
- JARVIS/CGCNN atom feature lookup instead of weak repeated atomic-number features.
- Recursive cutoff expansion, periodic edge canonicalization, and explicit reverse edges in graph construction.
- Validation and test loaders with `drop_last=False`, so all validation and test samples contribute to metrics.
- Seeded training and seeded data-loader shuffling for reproducibility.

- Target transforms for nonnegative/skewed targets, while reporting metrics after inverse transform on the original physical scale.
- Target-range-aware sample weighting, allowing low-target and high-target regions to be emphasized or de-emphasized directly.
- Optional nonnegative prediction clipping for targets that cannot be physically negative.

2.2 Band Gap: Failure Mode and Fix

For `optb88vdw_bandgap`, the test set was dominated by metallic or near-metallic samples, but the largest errors were also affected by underprediction of high-gap structures. A generic SmoothL1 run improved some tail metrics but did not directly optimize both regimes.

The improved objective therefore used:

- `SmoothL1` loss for robust regression.
- Extra weight for near-zero band gaps using `--low-target-threshold 0.05 --low-target-weight 2.0`.
- Stronger weight for high-gap samples using `--high-target-threshold 1.0 --high-positive-weight 4.0`.
- Nonnegative prediction clipping with `--prediction-min 0`.

This directly addressed the two observed band-gap error regimes: metallic samples and larger-gap underprediction.

2.3 Energy Above Hull: Failure Mode and Fix

For `ehull`, the imported original split was highly distribution-shifted. The train split contained many high-energy structures, while validation and test were mostly stable or near-stable structures. The target distributions were:

- Train mean: approximately `0.331`; about `26.6%` above `0.5`.
- Validation mean: approximately `0.041`; about `54.6%` below `0.001`.
- Test mean: approximately `0.0205`; about `75.5%` below `0.001`.

The original and early reimplementation runs had positive bias on stable/near-zero structures. The improved objective therefore used:

- `log1p` target transform for the nonnegative skewed target.
- Strong upweighting of near-zero structures using `--low-target-threshold 0.001 --low-target-weight 6.0`.
- Downweighting of positive and high-energy structures using `--positive-weight 0.35 --high-target-thre 0.1 --high-positive-weight 0.2`.
- Nonnegative prediction clipping with `--prediction-min 0`.

This reduced the near-zero positive bias from the earlier tuned runs and aligned training pressure with the validation/test distribution.

3 Results

Target	Run	MAE	RMSE	P95 Abs. Error	Wins/Losses
`optb88vdw_bandgap`	Original	0.26809	0.64872	1.47607	–
`optb88vdw_bandgap`	Improved	**0.23485**	**0.60921**	**1.19387**	**717/283**
`ehull`	Original	0.05503	0.08726	0.17663	–
`ehull`	Improved	**0.03213**	**0.07361**	**0.15353**	**788/212**

Table 1: Same-split test metrics on 1000 original test JIDs per target. Wins/losses count samples where the improved model had lower absolute error than the original model.

4 Reproduction Commands

All commands below are intended to be run on the remote instance from `$HOME/projects/alignn` with `uv` on the path.

4.1 Band Gap

```
export PATH="$HOME/.local/bin:$PATH"
cd "$HOME/projects/alignn" && uv run alignn alignn-train-small \
  --dataset dft_3d \
  --target optb88vdw_bandgap \
  --train-subset-size 0 \
  --val-subset-size 0 \
  --test-subset-size 0 \
  --batch-size 16 \
  --hidden-dim 128 \
  --alignn-layers 4 \
  --gcn-layers 4 \
  --epochs 25 \
  --seed 123 \
  --learning-rate 0.001 \
  --weight-decay 0.00001 \
  --loss smoothl1 \
  --scheduler onecycle \
  --target-transform none \
  --low-target-threshold 0.05 \
  --low-target-weight 2.0 \
  --high-target-threshold 1.0 \
  --high-positive-weight 4.0 \
  --prediction-min 0 \
  --run-name dft3d_optb88vdw_bandgap_h128_smoothl1_low2_high4_clamp_25epoch_seed123 \
  --project-root "$HOME/projects/alignn"
```

Result artifact paths:

- `results/checkpoints/dft3d_optb88vdw_bandgap_h128_smoothl1_low2_high4_clamp_25epoch_seed123.p`
- `results/logs/dft3d_optb88vdw_bandgap_h128_smoothl1_low2_high4_clamp_25epoch_seed123_history.c`
- `results/logs/dft3d_optb88vdw_bandgap_h128_smoothl1_low2_high4_clamp_25epoch_seed123_test_pred`

4.2 Energy Above Hull

```
export PATH="$HOME/.local/bin:$PATH"
cd "$HOME/projects/alignn" && uv run alignn alignn-train-small \
  --dataset dft_3d \
  --target ehull \
  --train-subset-size 0 \
  --val-subset-size 0 \
  --test-subset-size 0 \
  --batch-size 16 \
  --hidden-dim 128 \
  --alignn-layers 4 \
  --gcn-layers 4 \
  --epochs 25 \
  --seed 123 \
  --learning-rate 0.001 \
  --weight-decay 0.00001 \
  --loss smoothl1 \
  --scheduler onecycle \
  --target-transform log1p \
  --positive-weight 0.35 \
  --low-target-threshold 0.001 \
  --low-target-weight 6.0 \
  --high-target-threshold 0.1 \
  --high-positive-weight 0.2 \
  --prediction-min 0 \
  --run-name dft3d_ehull_h128_smoothl1_log1p_low6_downhigh_25epoch_seed123 \
  --project-root "$HOME/projects/alignn"
```

Result artifact paths:

- `results/checkpoints/dft3d_ehull_h128_smoothl1_log1p_low6_downhigh_25epoch_seed123.pt`
- `results/logs/dft3d_ehull_h128_smoothl1_log1p_low6_downhigh_25epoch_seed123_history.csv`
- `results/logs/dft3d_ehull_h128_smoothl1_log1p_low6_downhigh_25epoch_seed123_test_predictions.c`

5 Conclusion

The improvements came from target-specific error analysis rather than a generic architecture-only change. For band gap, weighting both metallic and high-gap regimes reduced aggregate error and tail error. For energy above hull, correcting the train/test distribution mismatch with near-zero-focused weighting removed most of the positive bias on stable structures. Both improved models outperform the original same-split test results on MAE, RMSE, p95 absolute error, and per-sample win/loss count.